

Task Management Platform DB Description

Entities:

- **Users:** Stores system users, credentials, and access levels (Admin, Member, Viewer).
- **Projects:** Defines project workspaces, created and managed by team members.
- **Tasks:** Core entity representing task cards, including Kanban status, priority, duration, JSON checklist subtasks, and board positioning.
- **Task_Assignees:** Pivot entity enabling many-to-many assignments between users and tasks.
- **Comments:** Contains discussion logs and comments posted by users within specific tasks.
- **Task_Dependencies:** Self-referencing link entity tracking task prerequisites for graph scheduling and cycle detection.
- **Notes:** Stores private user scratchpads for drafting personal ideas before task creation.

Relationships Summary

Relationship	Entity A	Cardinality A	Entity B	Cardinality B	Description
created_by	Users	Optional (1)	Projects	Mandatory (N)	A user can create zero or many projects; each project has one creator.
contains	Projects	Mandatory (1)	Tasks	Optional (N)	A project contains zero or many tasks; each task belongs to one project.
assigned_user	Users	Mandatory (1)	Task_Assignees	Optional (N)	Links users to their assigned tasks.
assigned_task	Tasks	Mandatory (1)	Task_Assignees	Optional (N)	Links tasks to their assigned users.
has_comments	Tasks	Mandatory (1)	Comments	Optional (N)	A task can have multiple user comments.
writes	Users	Mandatory (1)	Comments	Optional (N)	A user can author multiple comments.
owns_notes	Users	Mandatory (1)	Notes	Optional (N)	A user owns a personal scratchpad area.
is_dependent_on	Tasks	Mandatory (1)	Task_Dependencies	Optional (N)	Identifies the main blocked task (task_id).
blocks_task	Tasks	Mandatory (1)	Task_Dependencies	Optional (N)	Identifies the prerequisite parent task (depends_on_task_id).